

Rethinking Decoder Redundancy in Highly Efficient 3D Medical Image Segmentation

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Abstract

Efficient 3D segmentation decoders are commonly evaluated by aggregate accuracy and computational cost, leaving a basic question unanswered: where does additional decoder computation actually improve a prediction? We study this question through a controlled reimplementation of a full 3D UX-Net decoder and its compressed EffiDec3D counterpart on 13-organ abdominal CT segmentation. Rather than treating every disagreement as a benefit, we define voxel-wise decoder marginal utility using positive transitions (efficient wrong, full correct), negative transitions (efficient correct, full wrong), and their net difference. Our current study yields three consistent observations. First, errors are spatially heterogeneous: boundary error is $3.93\times$ interior error, and organ difficulty is negatively associated with organ size ($\rho = -0.544$). Second, the aggregate benefit of the full decoder is small and bidirectional; across two efficient-decoder seeds, positive and negative transition rates differ by only 0.065–0.130% of evaluated voxels. Third, this marginal benefit is predictable: subject-level entropy–net-utility correlations are 0.665 and 0.646 across the two seeds, while a preliminary voxel oracle places approximately 86% of positive transitions in the top 20% entropy-ranked foreground voxels. These findings motivate selective decoder allocation, but do not by themselves demonstrate realizable savings. We therefore introduce a paired, subject-bootstrap evaluation protocol based on net utility and contiguous 3D regions, separating oracle opportunity from deployable selection. [TODO: Replace the preliminary voxel-oracle sentence with corrected region-level O9 results and add cross-backbone/cross-dataset findings before submission.]

1. Introduction

Volumetric medical image segmentation relies on decoders to fuse multi-scale features and restore spatial detail. This restoration is expensive: every high-resolution 3D convo-

lution operates over millions of voxels, often with wide feature maps. Modern architectures therefore face a tension between anatomical accuracy and the memory, latency, and energy required for deployment. This tension has motivated efficient architectures and automated configuration pipelines such as nnU-Net [2], as well as transformer and large-kernel models including UNETR [1], Swin UNETR [5], and 3D UX-Net [3].

EffiDec3D [4] recently showed that decoder channels and high-resolution stages can be aggressively reduced while retaining competitive aggregate Dice. Its 3D UX-Net variant reports a 96.4% reduction in decoder parameters and a 93.0% reduction in decoder FLOPs. This is strong evidence that conventional decoders are over-provisioned in aggregate. Aggregate metrics, however, cannot reveal whether the remaining utility is uniformly distributed. A static decoder assigns the same architecture to every subject, organ, and location, although the need for detail restoration may vary sharply.

This distinction matters. A difficult or uncertain voxel is not necessarily a voxel that benefits from a stronger decoder: both models may fail there. Conversely, model disagreement is not necessarily improvement, because the full decoder can destroy a prediction that the efficient decoder got right. We therefore ask:

Where is the marginal utility of additional decoder capacity, how concentrated is it, and can it be identified without access to ground truth?

We answer this question with a controlled marginal-utility study. We compare a full 3D UX-Net decoder (Full-UX) with an EffiDec3D-compressed decoder (Effi-UX) under a common dataset, split, preprocessing pipeline, training schedule, and evaluation protocol. At each voxel we distinguish positive, negative, and net transitions between their predictions. We then analyze this utility by anatomy, boundary proximity, uncertainty, training stage, and selection budget. Statistical comparisons are performed at the subject level to avoid treating millions of correlated voxels

074 as independent samples.
 075 Our current contributions are:
 076 • We formulate *decoder marginal utility*, separating addi-
 077 tional decoder capacity from generic segmentation error
 078 and explicitly accounting for both corrected and damaged
 079 predictions.
 080 • We provide a controlled audit of full and efficient 3D
 081 UX-Net decoders across two efficient-decoder seeds, re-
 082 porting accuracy, HD95, parameters, MACs, latency, and
 083 memory together with the observed reproduction gap.
 084 • We characterize the anatomical and temporal distribution
 085 of decoder opportunity, including boundary/interior er-
 086 ror, organ size, uncertainty, and training-checkpoint evo-
 087 lution.
 088 • We define a selective-allocation protocol with predeclared
 089 budgets, paired subject bootstrap, net utility, and contigu-
 090 ous 3D regions. This protocol distinguishes a voxel oracle
 091 from a selector that could motivate an adaptive decoder.
 092 The study intentionally stops short of claiming an effi-
 093 cient adaptive method. Measured dynamic compute savings
 094 belong to a subsequent method study; here our goal is to es-
 095 tablish whether the empirical premise for such a method is
 096 sound.

097 2. Related Work

098 **3D medical image segmentation.** U-shaped encoder-
 099 decoder networks remain the dominant design for volumet-
 100 ric segmentation. UNETR [1] uses a transformer encoder
 101 and a convolutional decoder connected by multi-scale skips,
 102 while Swin UNETR [5] introduces hierarchical shifted-
 103 window representations. 3D UX-Net [3] instead approx-
 104 imates hierarchical transformer behavior with large-kernel
 105 depthwise convolutions. nnU-Net [2] demonstrates that
 106 data-dependent pipeline configuration and rigorous evalu-
 107 ation can be as important as architectural novelty. These
 108 works primarily optimize segmentation accuracy; our study
 109 instead examines the spatial utility of the decoder computa-
 110 tion already present in such systems.

111 **Efficient decoders.** EffiDec3D [4] observes that wide,
 112 high-resolution decoder stages dominate the cost of sev-
 113 eral 3D models. It uses a uniform reduced channel width
 114 and removes stages whose aggregate accuracy contribution
 115 is small. Across 12 tasks, the resulting static decoder pre-
 116 serves competitive accuracy at much lower cost. We take
 117 this result as a starting point and ask a different question:
 118 whether the small residual benefit of the full decoder is spa-
 119 tially uniform. Our work is an evaluation of decoder utility,
 120 not a new compression block.

121 **Uncertainty and selective computation.** Predictive en-
 122 tropy and confidence are widely used as uncertainty proxies.

They identify ambiguous predictions, but ambiguity and de-
 coder utility are not equivalent: an uncertain location may
 remain wrong after additional decoding, whereas a confi-
 dent error may still be correctable. Accordingly, we evalu-
 ate signals against held-out *net transitions* between matched
 decoders rather than against error alone. We also distin-
 guish arbitrary voxel ranking from contiguous region selec-
 tion, since sparse isolated voxels cannot be processed effi-
 ciently by standard 3D convolutions.

Evaluation and reproducibility. Small medical datasets
 create substantial seed and split variance, while voxel-level
 statistical tests can produce severe pseudo-replication. Our
 protocol therefore reports seed-level results, discloses de-
 viations from published baselines, and bootstraps subjects
 rather than voxels. This emphasis on auditing what aggre-
 gate metrics conceal aligns with the goals of the WACV
 Evaluations & Datasets track.

3. Marginal-Utility Study

3.1. Matched Decoder Comparison

Let f_E denote the efficient decoder and f_F the full decoder.
 For input volume x , their hard predictions at voxel v are
 $\hat{y}_E(v)$ and $\hat{y}_F(v)$, with reference label $y(v)$. The positive
 and negative transitions are

$$P(v) = \mathbb{I}[\hat{y}_E(v) \neq y(v)] \mathbb{I}[\hat{y}_F(v) = y(v)], \quad (1)$$

$$N(v) = \mathbb{I}[\hat{y}_E(v) = y(v)] \mathbb{I}[\hat{y}_F(v) \neq y(v)]. \quad (2)$$

We define discrete net marginal utility as

$$U(v) = P(v) - N(v). \quad (3)$$

This definition prevents every model disagreement from be-
 ing counted as an improvement. For a selected region S ,
 normalized net utility is

$$R_{\text{net}}(S) = \frac{\sum_{v \in S} U(v)}{\sum_v P(v)}. \quad (4)$$

The denominator makes positive recovery interpretable
 while allowing negative values when selection damages
 more predictions than it repairs. We report positive and neg-
 ative components alongside their difference.

3.2. Difficulty, Utility, and Predictability

We separate three maps that are often conflated: *error* in-
 dicates where Effi-UX is wrong; *uncertainty* is computed
 from its predictive distribution; and *utility* indicates where
 replacing its output with Full-UX changes correctness. Er-
 ror concentration alone does not establish an opportunity
 for additional decoding.

For Effi-UX class probabilities $p_c(v)$, predictive entropy is

$$H(v) = - \sum_c p_c(v) \log(p_c(v) + \epsilon). \quad (5)$$

We measure the association between entropy and net utility within each subject and aggregate subject-level coefficients with a nonparametric bootstrap. The final analysis will additionally report subject-level AUROC/AUPRC and calibration for predicting positive and net-beneficial regions.

3.3. Selective-Allocation Opportunity

We evaluate budgets $\mathcal{B} = \{5, 10, 20, 30, 50\}\%$ in two settings. First, entropy ranks individual foreground-union voxels, yielding a non-deployable oracle upper bound. Second, the volume is partitioned into 16^3 blocks, each scored by mean entropy. Selected blocks receive a four-voxel context halo; the halo-expanded volume is recorded as an executed volume proxy. Utility is evaluated on each selected core, reflecting a patch-based refiner that consumes context but writes only its core prediction.

At each budget, entropy selection is compared with 100 matched random selections. Confidence is a second zero-forward-pass baseline. The primary budget is declared as 20%; 10% and 30% are sensitivity analyses. A paired bootstrap resamples the same subject indices for entropy and random policies. The opportunity criterion is met only if mean region-level net utility is positive and the lower 95% confidence bound for entropy minus random exceeds zero at the primary budget.

3.4. Anatomical and Training Analyses

We compare errors at a three-voxel foreground boundary band and eroded interior. Per-organ Dice, entropy, volume, and error characterize anatomical heterogeneity. Spearman correlation measures the relation between organ size and difficulty; a partial correlation tests whether entropy explains difficulty beyond log organ volume. Finally, checkpoints at 5k, 10k, 20k, 30k, and 45k iterations measure how uncertainty changes during training.

4. Experimental Protocol

Dataset and split. We use the 13-organ BTCV/Synapse abdominal CT task with 18 training and 12 validation volumes. The evaluated organs are spleen, kidneys, gallbladder, esophagus, liver, stomach, aorta, inferior vena cava, portal/splenic veins, pancreas, and adrenal glands. Images are intensity-windowed and resampled according to the public EffiDec3D training pipeline; training uses 96^3 patches. The same validation subjects are paired across all model comparisons.

Models. Full-UX is the public full 3D UX-Net configuration with feature widths [48, 96, 192, 384]. Effi-UX uses the

Table 1. BTCV controlled reimplement. Published values are shown only for calibration; our paired analyses use the local runs. MAC denotes multiply-accumulate operations under our profiler.

Model	Dice $\uparrow$	HD95 $\downarrow$	MAC (G)	Params (M)	Lat. (ms)
Published Full-UX	.7974	–	631.97	53.00	–
Published Effi-UX	.7925	10.12	51.47	3.16	–
Full-UX, seed 0	.7918	9.04	578.74	53.007	40.6
Effi-UX, seed 0	.7700	14.41	41.06	2.955	7.5
Effi-UX, seed 1	.7755	10.17	41.06	2.955	7.6

corresponding EffiDec3D decoder with 48 channels, additive skip aggregation, and resolution factor two. Both are trained for 45k iterations with Dice-cross-entropy loss and AdamW. We evaluate one Full-UX run and two independently initialized Effi-UX runs. [TODO: Add a matched second Full-UX seed or explain why only E1 seed sensitivity is used.]

Evaluation. Segmentation quality is reported using mean and per-organ Dice and 95th percentile Hausdorff distance (HD95). Efficiency measurements include parameters, MACs, latency, peak inference memory, and training time. Sliding-window inference uses a 96^3 region, batch size four, overlap 0.7, and constant blending. Observation confidence intervals resample subjects.

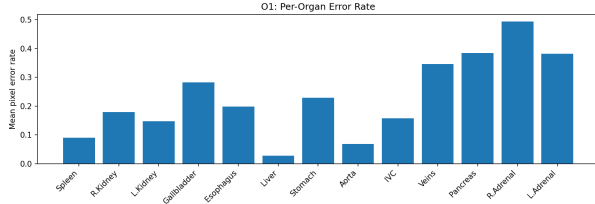
Reimplementation status. Our data were reconstructed from the public TransUNet/Synapse release, whose NIfTI files have identity affine metadata, whereas the EffiDec3D paper refers to a processed `btcv_trns` dataset. Consequently, we present the experiment as a *controlled reimplementation*, not an exact reproduction. All comparisons within our study use the same local data and evaluation code. We report the discrepancy from published numbers in full.

Planned generalization. The final submission will repeat the utility analysis on a second backbone (Swin UNETR or MedNeXt) and a second dataset/modality (FeTA MRI). [TODO: Insert O7 cross-dataset and O8 cross-backbone protocols/results.]

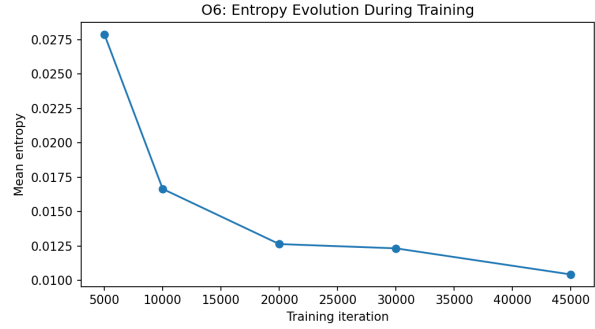
5. Results

5.1. Controlled Reimplementation

Table 1 shows that the full model closely approaches the published Dice (a 0.56-point difference), whereas the two efficient runs remain 1.70–2.25 points below the published efficient model. Seed 1 nearly matches the published HD95. The two-seed efficient mean Dice is 0.7728. Accordingly,



(a) Per-organ error.



(b) Entropy over training.

Figure 1. Current anatomical and temporal observations for Effi-UX seed 1. Entropy contracts during training but a residual difficult subset remains.

seed variance explains part, but not all, of the reproduction gap. The local E0–E1 Dice gap (1.63–2.18 points) is larger than the published 0.49-point gap, and we do not attribute its full magnitude to decoder compression.

Efficiency trends reproduce strongly: Effi-UX uses $14.1\times$ fewer MACs, $17.9\times$ fewer parameters, $5.3\text{--}5.4\times$ lower measured latency, and approximately $7.7\times$ lower peak inference memory than our Full-UX.

5.2. Errors Are Anatomically Concentrated

For seed 1, mean boundary error is 0.1892 versus 0.0481 in the eroded interior, a ratio of 3.93. Small and thin structures are disproportionately difficult: the organ-size/difficulty Spearman correlation is -0.544 . Entropy retains substantial association with organ difficulty after controlling for log volume (partial $r = 0.810$), suggesting that size alone does not explain the signal.

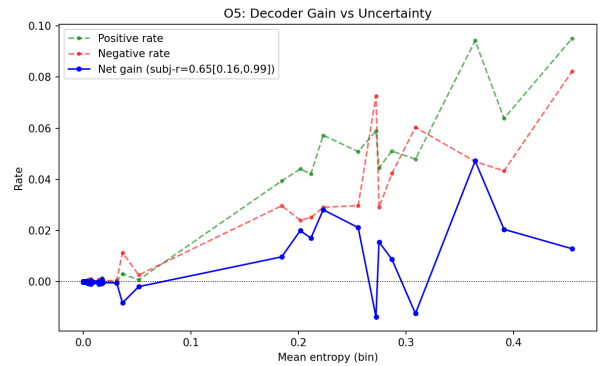
Difficulty also evolves systematically. Mean entropy decreases from 0.0279 at 5k iterations to 0.0166, 0.0126, 0.0123, and 0.0104 at 10k, 20k, 30k, and 45k, respectively (Fig. 1). Training removes much of the global ambiguity but does not eliminate the concentrated residual.

5.3. Decoder Utility Is Small and Bidirectional

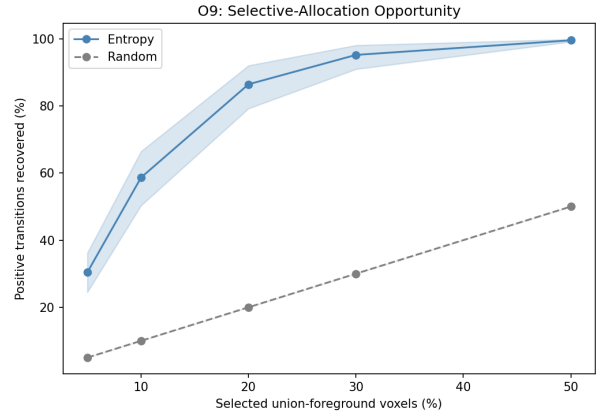
The full decoder is not uniformly better. For seed 0 it repairs approximately 0.339% of voxels while damaging 0.209%; for seed 1 the corresponding values are 0.290% and 0.225%. Thus the net label-transition benefit is only 0.065–0.130%

Table 2. Marginal-utility observations across efficient-decoder seeds. Rates are fractions of evaluated voxels. O9 is a preliminary voxel oracle and is not the final contiguous-region result.

Metric	E1 seed 0	E1 seed 1
Positive transition rate	.00339	.00290
Negative transition rate	.00209	.00225
Net transition rate	.00130	.00065
Entropy–net utility r	.665	.646
Bootstrap 95% CI	[.171,.996]	[.158,.994]
Top-20% positive recovery	.864	.864



(a) Net gain vs. entropy.



(b) Legacy positive-only oracle.

Figure 2. Preliminary predictability results for seed 1. Panel (b) ranks individual voxels and counts only positive transitions; it is shown to motivate, not replace, the corrected contiguous-region analysis.

of voxels. Although this label-level statistic is not identical to Dice, it shows why positive and negative transitions must be reported together.

5.4. Utility Is Predictable, but Current O9 Is an Upper Bound

The subject-level entropy-net-utility correlation is positive for both efficient seeds (Table 2), and its subject-bootstrap lower bound remains above zero. The legacy voxel analysis finds that the top 20% entropy-ranked foreground-union voxels contain about 86.4% of positive transitions in both runs. However, isolated voxels cannot be processed efficiently by standard 3D convolutions, and positive-only recovery ignores negative transitions. We therefore treat this result strictly as an oracle upper bound.

[TODO: Replace Fig. 2(b) and the final paragraph with O9_corrected: block-level positive, negative, and net utility; actual halo-expanded volume; matched-random mean; and paired 95% CI for both seeds.]

5.5. Routing Signals

On seed 1, confidence and entropy have similar correlations with net utility (0.663 and 0.655, respectively). Monte-Carlo dropout is not a valid comparator for this checkpoint because the evaluated architecture has no active dropout; its measured stochastic variance is effectively zero. We therefore retain confidence as a required cheap baseline and do not claim entropy is uniquely optimal.

6. Discussion and Limitations

What the evidence supports. Across two efficient-decoder seeds, the direction of the central observation is stable: additional decoder capacity produces a small, bidirectional aggregate effect, and its net benefit increases with an uncertainty proxy. Boundary and organ analyses further show that segmentation difficulty is spatially heterogeneous. Together, these findings justify testing selective regional decoding rather than applying a uniformly expensive decoder.

What it does not yet support. The current study does not demonstrate lower *executed* compute for an adaptive model. The legacy top-20% result is a voxel oracle, not a realizable router. Even a block opportunity curve uses full-model predictions only to define retrospective utility; it does not account for errors introduced by a learned refinement module. Dynamic FLOPs, latency, memory, and final Dice must therefore be evaluated in a separate method study.

Reproduction gap. Our efficient baseline remains 1.70–2.25 Dice points below the published result, while the full baseline is much closer. The most plausible unresolved difference is data preprocessing: our reconstructed files have identity affine metadata, whereas the original work used a processed dataset not distributed with the repository. The enlarged local full-efficient gap could inflate the amount of

apparent decoder opportunity. We mitigate this by disclosing all values, analyzing two efficient seeds, and limiting claims to controlled local comparisons. A paper-equivalent preprocessing run and matched multi-seed full baseline remain important controls.

Statistical limitations. The dataset contains only 12 validation subjects. Confidence intervals are therefore wide, and millions of voxels must not be interpreted as independent observations. Our early entropy-error coefficient pooled bins across subjects and is retained only as a descriptive diagnostic. Final error predictability will use subject-level AUROC/AUPRC and calibration. Budgets and primary endpoints must be declared before examining the corrected O9 results.

Generalization. All completed results currently use one CT dataset and one backbone family. Without cross-backbone and cross-modality replication, the finding may be specific to 3D UX-Net, the Synapse split, or EffiDec3D’s resolution restriction. The final claim will be narrowed if O7/O8 do not reproduce the direction.

Clinical interpretation. High uncertainty and small-organ difficulty do not imply clinical safety. Entropy can miss confidently wrong predictions, and improving average Dice does not guarantee preservation of clinically important boundaries. Any future adaptive decoder should therefore retain a conservative fallback and be evaluated per organ and per subject, not only by average compute.

7. Conclusion

We presented a marginal-utility perspective on efficient 3D segmentation decoders. A controlled full-versus-efficient comparison shows that additional decoder capacity has a small and bidirectional aggregate effect, while its remaining utility is associated with a spatially concentrated subset of difficult regions. These trends are stable across two efficient-decoder seeds, despite an unresolved absolute reproduction gap. More broadly, our results show why aggregate Dice/FLOPs tables are insufficient for reasoning about decoder redundancy: error, uncertainty, and decoder-specific utility are distinct quantities. The proposed net-utility, paired-bootstrap, and contiguous-region protocol provides a stricter test of whether selective allocation is empirically justified.

[TODO: Update the conclusion after corrected O9 and O7/O8. If region-level net utility or generalization fails, narrow the claim to heterogeneity and report the negative result rather than retaining “predictable.”]

References

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